I designed a regression comparison and diagnosis experiment using patients' basic physical indicators to predict the degree of disease progression one year later. The diabetes regression data set built in scikit-learn is used. This dataset has 442 samples with 10 input features, including patient age, gender, BMI, and so on.

I chose this dataset because its continuous target and ten numerical features make it suitable for comparing regression models. It is also included in scikit-learn, making the experiment easy to reproduce.

The data were split into 80% training and 20% testing. Five-fold cross-validation on the training set compared four models: a mean baseline, ordinary linear regression, Ridge, and Lasso.

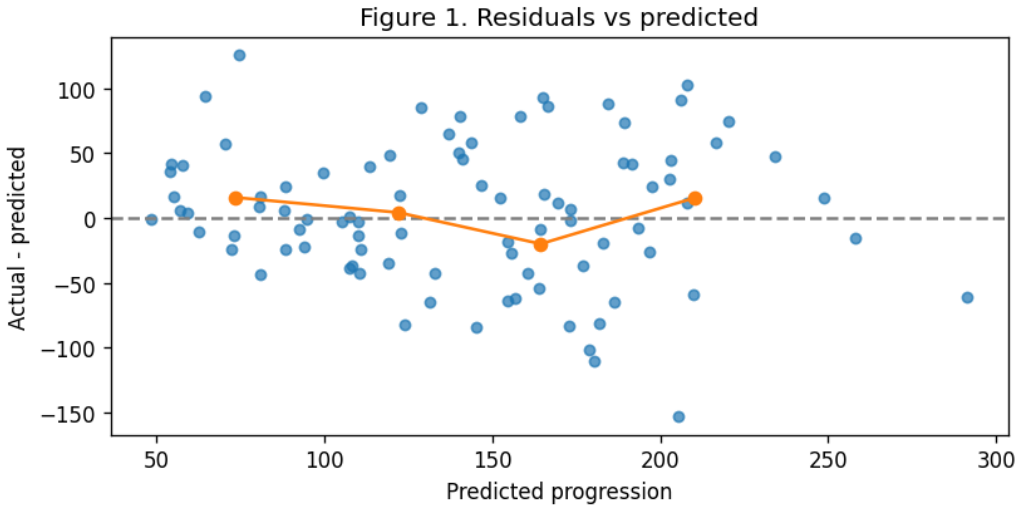
The main research questions were how much these models improve on predicting the mean and whether regularization improves linear regression. I also examined residuals, Cook's distance, learning curves, and errors across BMI groups.

Results

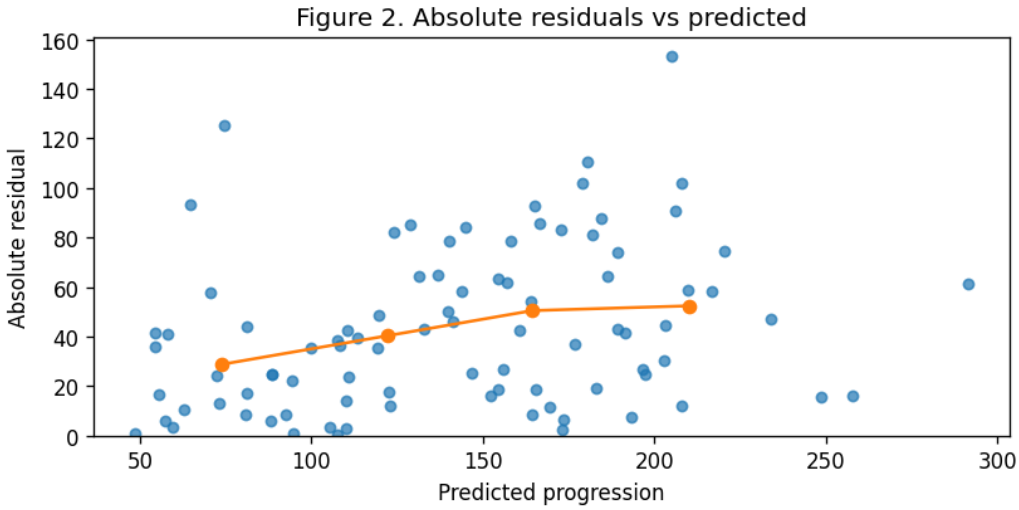
Model	Alpha	CV MAE	Test MAE	RMSE	R2
Dummy	-	66.867	64.006	73.222	-0.012
Linear	-	45.048	42.794	53.853	0.453
Ridge	1	44.999	42.812	53.777	0.454
Lasso	0.1	44.956	42.805	53.709	0.456

Lasso (alpha = 0.1) had the lowest CV MAE, 44.956. Ridge and Lasso improved CV MAE over ordinary linear regression by only 0.049 and 0.093, so regularization provided little benefit.

Residual diagnostics



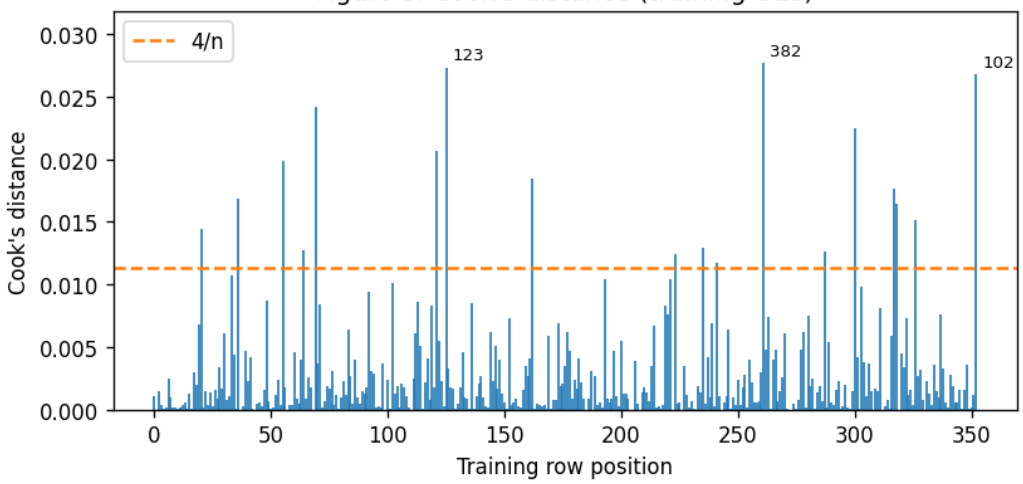
Residuals show local bias, including overprediction in the third prediction quartile (mean residual -20.02).



Average absolute errors increase with predicted values, suggesting possible heteroscedasticity.

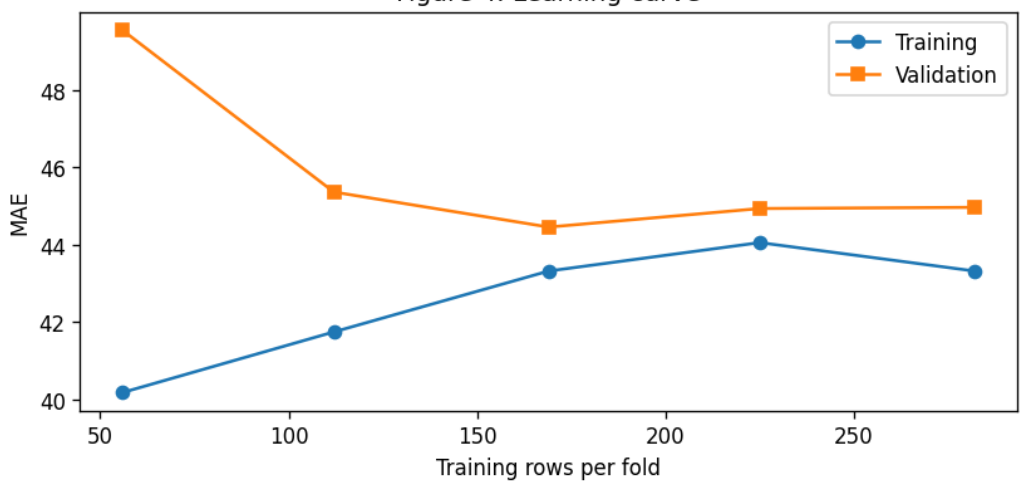
Influence and learning curve

Figure 3. Cook's distance (training OLS)



Eighteen training rows exceeded the Cook's distance screening threshold (4/n); row 382 had the largest value. No rows were removed.

Figure 4. Learning curve



Validation MAE decreased from 49.53 to 44.96 and then leveled off. The final train–validation gap was only 1.64.

Test MAE was 34.490 for low BMI (n = 38), 47.433 for middle BMI (n = 23), and 50.288 for high BMI (n = 28).

The three largest test errors were:

Row	Actual	Predicted	Absolute error	BMI
56	52.000	205.195	153.195	30.2
77	200.000	74.619	125.381	23.0
211	70.000	180.323	110.323	29.8

A small dataset and single split limit generalization. Coefficients describe associations rather than causal effects.